

JEE ADVANCED WEEKEND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



Sec: SR.IIT_*CO-SC(MODEL-A)
Time: 3 Hrs

Date: 23-06-24
Max. Marks: 198

23-06-24_SR.IIT_STAR CO-SC(MODEL-A)_JEE ADV_CAT-33_SYLLABUS

MATHEMATICS: **DESIGNATED TOPIC (100%):** Application of Derivatives

PHYSICS: **DESIGNATED TOPIC (100%):** Elasticity, Viscosity, units and dimensions, Error analysis, vernier calipers & Screw gauge

CHEMISTRY: **DESIGNATED TOPIC(100%):** Phenol, Ether, Chemical kinetics, Group-16

THE NARAYANA GROUP

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 13 – 18)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				18	66

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 24)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 31 – 36)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				18	66

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 42)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 43 – 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 49 – 54)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				18	66

THE NARAYANA GROUP

MATHEMATICS**MAX.MARKS: 66****SECTION – 1 (Maximum Marks: 18)**

This section contains SIX (06) questions.

The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9 , BOTH INCLUSIVE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric

keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

1. The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for $x = k$, the area of the triangle is maximum, then $3(k^2 - 1)$ is equal to _____
2. The number of distinct real roots of the equation $x^7 - 7x - 2 = 0$ is _____
3. The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is _____
4. If the interval of x for which the function $y = -x(x - 2)^2$ increases is given by $\left(\frac{2}{3}, \lambda\right)$ then find maximum value of λ
5. The value of 'c' in Lagrange's mean value theorem for $f(x) = x^3 - 2x^2 - x + 4$ in $[0, 1]$ is given by $\frac{1}{\lambda}$ then find λ .
6. Consider a point P_1 on the curve $y = x^3$ such that the tangent at $P_1(1,1)$ meets the curve again at P_2 . And the tangent at P_2 meets the curve at P_3 and so on.

Let (x_n, y_n) be the coordinates of P_n then the value of $\frac{\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{x_r}}{\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{y_r}}$ is equal to $\frac{m}{n}$, where m and n are co-prime positive integers, find the value of $(m + n)$

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

7. A channel 27 m wide falls at a right angle into another channel 64 m wide, the length of the log that can be floated along this system of channels is l . Then l can be
- A)120 B) 125 C)150 D)144
8. Let $f(x) = (x-1)^4 \cdot (x-2)^n, n \in N$. Then $f(x)$ has
- A) a local maximum at $x = 1$ if n is odd
- B) a local maximum at $x = 1$ if n is even
- C) a local minimum at $x = 2$ if n is even
- D) a local maximum at $x = 2$ if n is odd
9. Let $f(x) = x \cdot \sin \pi x, x > 0$. Then for all natural numbers $n, f'(x)$ vanishes at
- A) A unique point in the interval $\left(n, n + \frac{1}{2}\right)$
- B) A unique point in the interval $\left(n + \frac{1}{2}, n + 1\right)$
- C) A unique point in the interval $(n, n + 1)$
- D) Two points in the interval $(n, n + 1)$

10. Let $f : R \rightarrow R$ be a continuous function such that $f(1) = 2, f(4) = 8$. If $f'(x) \leq 2 \forall x \in R$, then which of the following is always true?
- A) $f(x)$ is increasing in $[1, 4]$
 - B) $f(x)$ is increasing in R
 - C) There exists atleast three values of $x_0 \in R$ such that $f''(x_0) = 0$
 - D) $f(0) \geq 0$
11. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x)]$; where $[\cdot]$ is greatest integer function and $f(x)$ is non-constant continuous function, then
- A) $f(x)$ is increasing at $x = a$
 - B) $f(a)$ is an integer
 - C) $f(x)$ is having local maximum at $x = a$
 - D) $f(x)$ is having local minimum at $x = a$
12. Which of the following statements are always true about a function $f : R \rightarrow R$?
- A) If $f(x)$ is invertible then $f(x)$ has no local extremum points
 - B) If a tangent can be drawn to the curve $y = f(x)$ at $x = a$, then $f(x)$ is differentiable at $x = a$
 - C) If $f(x)$ is a bijection then $f(x) = 0$ has exactly one root
 - D) If $f(x)$ is differentiable at a point then it is continuous at that point.

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

13. Let $f(x)$ be a cubic polynomial with $f(1) = -10$, $f(-1) = 6$, and has a local minima at $x = 1$, and $f'(x)$ has a local minima at $x = -1$. Then $f(3)$ is equal to _____
14. A man starts walking from the point $P(-3, 4)$ touches the x -axis at R , and then turns to reach at the point $Q(0, 2)$. The man is walking at a constant speed. If the man reaches the point Q in the minimum time, then $50((PR)^2 + (RQ)^2)$ is equal to _____
15. Water is being filled at the rate of $1 \text{ cm}^3 / \text{sec}$ in a right circular conical vessel (vertex downwards) of height 35 cm and diameter 14 cm. When the height of the water level is 10 cm, the rate (in cm^2 / sec) at which the wet conical surface area of the vessel increases is λ , then $\lambda^2 =$ _____
16. Let f be a function defined on $\mathbb{R}$ (the set of all real numbers) such that $f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4$, for all $x \in \mathbb{R}$. If g is a function defined on $\mathbb{R}$ with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, for all $x \in \mathbb{R}$, then the number of points in $\mathbb{R}$ at which g has a local maximum is _____
17. For a polynomial $g(x)$ with real coefficients, let m_g denote the number of distinct real roots of $g(x)$. Suppose S is the set of polynomials with real coefficients defined by $S = \{(x^2 - 1)^2(a_0 + a_1x + a_2x^2 + a_3x^3) : a_0, a_1, a_2, a_3 \in \mathbb{R}\}$.
For a polynomial f , let f' and f'' denote its first and second order derivatives, respectively. Then the minimum possible value of $(m_{f'} + m_{f''})$, where $f \in S$, is _____
18. Let the function $f : (0, \pi) \rightarrow \mathbb{R}$ be defined by $f(\theta) = (\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^4$. Suppose the function f has a local minimum at θ precisely when $\theta \in \{\lambda_1\pi, \lambda_2\pi, \dots, \lambda_r\pi\}$, where $0 < \lambda_1 < \dots < \lambda_r < 1$. Then the value of $\lambda_1 + \dots + \lambda_r$ is _____

PHYSICS

MAX.MARKS: 66

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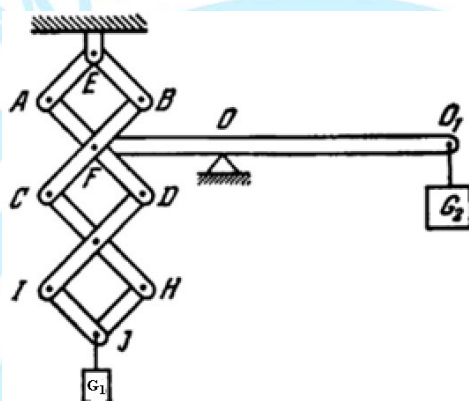
Answer to each question will be evaluated according to the following marking scheme:

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Negative Marks: -1 In all other cases.

19. Find the ratio between the weights G_1 and G_2 if the system shown in Fig. is in equilibrium Bars AD, BC, CH, DI and arm OO_1 of the lever are twice as long as bars AE, EB, IJ, JH and arm FO, respectively. Disregard the weight of the bars and the arm.



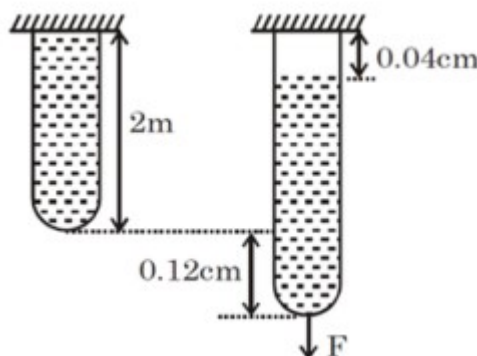
Now suppose the wire connecting G_1 (i.e. the load on the left side) has $3/2$ times the length of wire connecting G_2 , find the ratio of elongations in the two wires say I_1 / I_2 if the wires have same material and identical cross sectional areas. Report the ratio of the two elongations (i.e. I_1 / I_2) in your response sheet.

20. The nut on the bolt shown is rotated till the nut is just taut. The washer has the same area as that of the nut. Its thickness is 1 mm and area is 10cm^2 . The nut and bolt are perfectly rigid. The pitch of the screw is also 1 mm. After the just taut position, the nut is tightened by rotating it by 1.8° . If the stress in the washer is $2 \times 10^7 \text{ N/m}^2$, find the young's modulus Y (in N/m^2) for the washer. Fill $Y/10^9$ in OMR sheet. [The figure here is only representative and not to the scale. The washer is completely overlapped by the nut and the bolt head in the tightened position]



21. A uniform glass tube of length 2m is closed at the lower end. It is completely filled with water and rigidly fixed at upper end. Now the tube is stretched downwards. Although the length of tube increases by 0.12 cm, the length of water column increase only by 0.08 cm. (Assume water is incompressible)

What is poisson's ratio σ for glass. Fill $\frac{1}{\sigma}$ in OMR sheet.



[Assume that the support at the top allows free lateral contraction of the tube while maintaining the longitudinal stress, also ignore any and effects arising due to hemispherical curved surface at the bottom, i.e. the tube may be assumed to be practically cylindrical]

22. A certain gas obeys the equation of state $U(S, V, N) = aS^7 / V^4 N^2$, where a is a dimensioned constant. Here U represents the internal energy of the gas, S the entropy, V the volume and N the fixed number of particles of the system.

Let such a gas be filled in a box of volume V and the internal energy of the system be U . A partition is placed to divide the box into two each equal parts, each having volume $V/2$. For each part, the internal energy is now αU and the dimensioned constant be βa . Obtain α and β .

Now report the value of $4(\alpha + \beta)$

[Note that Entropy is an extensive property]

23. From classical electrodynamics, we know that an accelerated electron radiates energy. The expression for the power P of this radiation is given by

$$P = K \epsilon_0^w e^x a^y c^z$$

Where a is the acceleration, c is the speed of light, ϵ_0 is the permittivity of free space, and K is a dimensionless constant.

Obtain $\{w, x, y, z\}$ using dimensional analysis

Now report the value of $x + y + w + z$

24. The zero of main scale matches with zero of Vernier scale. 9th division of main scale matches with 10th division of Vernier. No other match is present. After sliding, zero of Vernier is found to be in between 3th & 4th division of main scale & 5th division of Vernier matches with xth division of main scale. The value of x is _____

{Given : 10VSD = 9MSD }

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Negative Marks: -2 In all other cases.

25. Prof. Saha gave the following problem to four students.

In this problem work done by a system on its surroundings is taken as positive. A non – ideal gas follows the Van der Waals equation of state

$$\left(P + \frac{n^2 a}{V^2} \right) (V - nb) = nRT$$

Where P, V, and T denote the pressure, volume, and temperature, respectively; n is the number of moles; R is the universal gas constant and a, b are dimensional positive constants. This gas expands adiabatically from an initial temperature T_i and volume V_i to a final temperature T_f and volume V_f . The adiabatic process is described by an equation of the form $f(P, V; n, a, b, \alpha) = \text{constant}$, where α is a dimensionless number which is greater than 1. It is given that $\alpha \rightarrow \gamma$ in the ideal gas limit, where γ is the adiabatic exponent. What is the work (W) done by the gas in the process?

The four students solved the problem independently and gave four different answers. Their answers were:

A) $W = \frac{nR}{\alpha - 1} (T_i - T_f) + n^2 a (V_f^{-1} - V_i^{-1})$

B) $W = \frac{nR}{\alpha - 1} (T_i - T_f) + n^2 a (V_f^{\alpha-1} - V_i^{\alpha-1})$

C) $W = \frac{nR}{\alpha - 1} (T_f - T_i) + n^2 a (V_f^{-1} - V_i^{-1})$

D) $W = \frac{nR}{\alpha - 1} (T_i - T_f) \left[1 - \left(\frac{V_f - nb}{V_i - nb} \right)^{\alpha-1} \right]$

Now, Prof. Saha had actually provided the exact expression of $f(P, V; n, a, b, \alpha)$ to the students, but could not remember it during evaluation. Still, he could determine that some or all of the four answers above must be incorrect, based on general physical arguments alone.

Consider each of the four answers and think at least one reason for each of them showing why it is wrong, or possibly correct. Note that you are not required to derive a correct expression for W or a detailed derivation for it in this question.

Now mark the definitely incorrect expression(s) for work

[You may assume that in Adiabatic Expansion, final temperature is always less than initial temperature]

26. There is a specially designed vernier calipers in which 4 main scale division are equal to 10 vernier scale division. In the zero error, 5th and 10th division of vernier scales coincides with 1mm and 3 mm mark on scale respectively and zero of the vernier scale is left to the zero of the main scale. During taking measurement 2nd and 7th divisions of the vernier scale coincide with 23mm and 25mm mark on main scale respectively.

(1 main scale division = 1 mm)

- A) Least count of this vernier calipers is 0.6 mm
 B) Instrument has negative zero error of 1mm
 C) Measurement of the length is 22.2 mm
 D) Measurement of the length is 23.2 mm
27. Area of the cross – section of a wire is measured using a screw gauge. The size of the main scale division is 0.5 mm. The circular scale has 100 divisions and for one full rotation of the circular scale, the main scale shifts by two divisions, The measured readings are listed below.

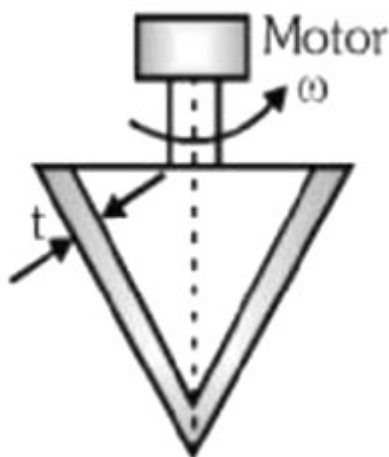
Measurement condition	Main scale reading	Circular scale reading
Two arms of gauge touching each other without wire	0 division	4 division
Attempt-1: With wire	4 divisions	20 divisions
Attempt-2: With wire	4 divisions	16 divisions

What are the diameter and cross – sectional area of the wire measured using the screw gauge?

A) $2.22 \pm 0.02 \text{ mm}, \pi(1.23 \pm 0.02) \text{ mm}^2$ B) $2.22 \pm 0.01 \text{ mm}, \pi(1.23 \pm 0.01) \text{ mm}^2$

C) $2.14 \pm 0.02 \text{ mm}, \pi(1.14 \pm 0.02) \text{ mm}^2$ D) $2.14 \pm 0.01 \text{ mm}, \pi(1.14 \pm 0.01) \text{ mm}^2$

28. In the shown figure, there is a conical shaft rotating on a bearing of very small clearance t . The space between the conical shaft and the bearing, is filled with a viscous fluid having coefficient of viscosity η . The shaft is having radius R and height h . If the external torque applied by the motor is τ and the power delivered by the motor is P working in 100% efficiency to rotate the shaft with constant ω then



A) $P = \frac{\pi \omega^2 \eta R^3 \sqrt{R^2 + h^2}}{2t}$

B) $\tau = \frac{\pi \omega \eta R^3 \sqrt{R^2 + h^2}}{2t}$

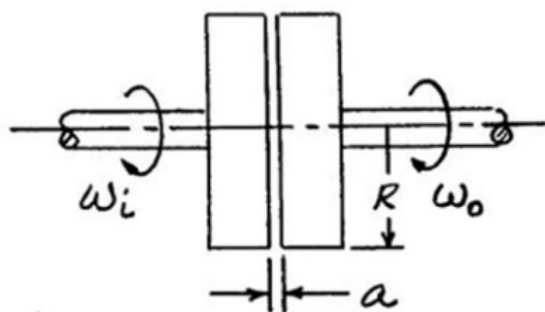
C) $P = \frac{\pi \omega^2 \eta R^2 h}{2t}$

D) $\tau = \frac{\pi \omega \eta^2 R^3 h^2}{2t \sqrt{R^2 + h^2}}$

29. Choose the correct statement:

- A) The viscosity of liquids in general tends to decrease with increase in temperature
- B) Viscosity of Gases in general tends to increase with increase in temperature
- C) All liquids obey Newton's law of viscosity
- D) The fluids such as Quick Sand, Tooth paste and Boot Polish do not obey Newton's law for viscosity and are hence classified as Non – Newtonian Fluids

30. A viscous clutch as shown in figure transmits torque. Radius of each clutch plate is R and separation between the plates is 'a' and is completely filled with liquid of coefficient of viscosity η . If ω_i and ω_o are angular velocities of plates connected to input and output respectively.



[Here we are applying some torque on the input shaft and getting some torque on the output shaft and in steady state the angular velocities of the two shafts are constant with time. The transmission efficiency is defined as the ratio of power supplied to the output shaft to the power supplied to the input shaft]

Now choose the correct statement:

- A) Torque transmitted is $\frac{\pi\eta(\omega_i - \omega_o)R^4}{2a}$
- B) Torque transmitted is $\frac{\pi\eta(\omega_i^2 - \omega_o^2)R^4}{2\omega_o a}$
- C) Transmission efficiency is $\frac{\omega_o}{\omega_i}$
- D) Transmission efficiency is $\frac{\omega_o^2}{\omega_i^2}$

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

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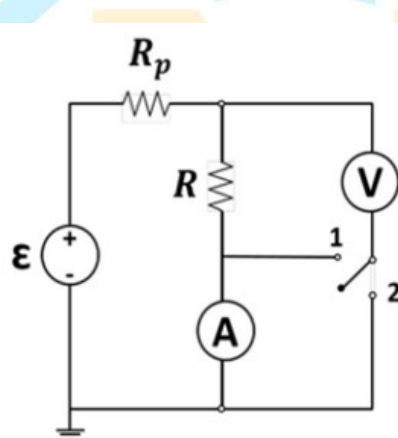
Zero Marks: 0 In all other cases.

31. A uniform rod is rotating about one of its end with a constant angular velocity ω in gravity free space. The ratio of strain produced in two halves is k ($k > 1$). The value of k is _____

32. The ammeter – voltmeter method is widely used for measuring electrical resistance in the physics laboratory. In this method, the resistance R is always derived from the readings V and I from a voltmeter and an ammeter respectively, using Ohm's law : $R = V / I$. While using this method, it is assumed that the ammeter and voltmeter used in the setup are ideal. In this problem, we will find the pitfalls of this assumption and devise a new setup with a better performance.

The standard ammeter – voltmeter setup consists of a DC voltage source (ϵ) maintained at a constant voltage, a protection resistance (R_p), an ammeter (A), and a voltmeter (V). The unknown internal resistance of the ammeter and the voltmeter are R_A and R_V , respectively. Also, $R_V \gg R_A$.

We aim to measure the true value R of an unknown resistor.

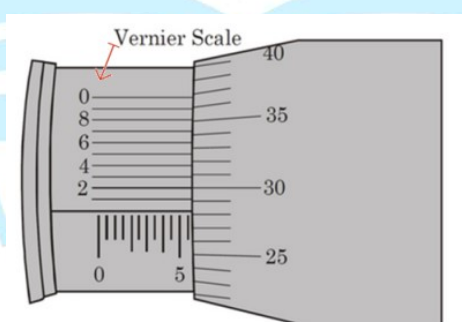


We consider a two commonly used circuit configurations (1) and (2) indicated by the two possible positions of the switch in the circuit diagram shown below. Let the measured values of the resistance R be R_{m1} and R_{m2} in the setup (1) and (2), respectively. The relative error, Δ , is defined as the ratio of the absolute error of the measurement to the actual value: $\Delta = (R_m - R) / R$

Obtain in relative errors in the measurements (Δ_1 and Δ_2) for each of the above configurations.

Now report the value of $1000(\Delta_1 + \Delta_2)$ if $R = 50\Omega$, $R_V = 3000\Omega$ and $R_A = 1\Omega$.

33. Pitch of a screw gauge is 1 mm and there are 100 divisions on its circular scale. When nothing is put between the studs, the zero of the main scale is not visible and 90^{th} circular scale division coincides with main scale. When circular scale is rotated by one complete rotation, zero is still not visible, but during the second rotation zero becomes just visible and when this happens, 10^{th} division of circular scale is coinciding with main scale. When a glass plate is placed between the studs, the circular scale lies between 9^{th} and 10^{th} division of main scale and 4^{th} division of circular scale coincides with main scale. If the thickness of plate in mm is x report the value of $100x - 115$
34. The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm & there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm & that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2%, the relative percentage error in the density is _____.
35. The picture shown below represents a vernier screw gauge. Main scale or pitch is 0.5 mm. There are 50 divisions on circular scale. 9 circular scale divisions correspond to 10 vernier scale divisions. What is the reading R (in mm)? Fill $10R$ in the OMR.
- [Assume no zero error]
- [A vernier screw gauge combines the concept of a screw gauge and a vernier scale to get extra precision. Here the reading of circular is refined to an extra decimal base on matching division on vernier scale]



36. A solid sphere of mass m and radius r is suspended by an elastic string of force constant k in a liquid. The coefficient of viscosity of the liquid is η . The angular frequency of its damped oscillations is found to be 4 rad/s. Find the reciprocal of η in SI units. Given, in SI units, $m = 1$, $k = 25$, and $r = 1$.

CHEMISTRY

MAX.MARKS: 66

SECTION – 1 (Maximum Marks: 18)

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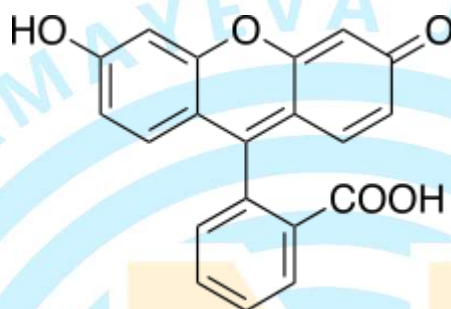
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

37. Eosin is formed by a reaction between bromine and fluorescein. The structure is given below



Fluorescein is an orange color dye used in medicine, it's brominated compound (called eosin) has enhanced applications as it is fluorescent and binds to specific proteins.

Number of 'Br' atoms in eosin molecule are _____

38. When phenol is nitrated using conc HNO_3 for prolonged time picric acid is obtained as a minor product and another product 'P' obtained as major product. Degree of unsaturation in the product 'P' is
39. Two substances 'A' and 'B' are initially present as $[A_0] = 8[B_0]$ and $t_{1/2}$ for the first order decomposition of 'A' and 'B' are 20 and 40 seconds, respectively. If they start decomposing at the same time, after how much time in minutes, the concentration of both of them would be same?
40. A substance is reduced to one third of its origin concentration in 100 minutes by first-order kinetics. The time (in min) in which it will be reduced to one ninth of its original value is 'X' then the value of $\frac{X}{40}$ is
41. If excess water is added into a 10g of Oleum bottle sample labelled as "112% H_2SO_4 " and is reacted with 5g Na_2CO_3 , then find the weight of CO_2 evolved at room temperature.
42. Number of S-S linkages present in sodium salt of hexathionic acid are

SECTION - 2 (Maximum Marks : 24)

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For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

43. A sulphate of a metal (A) on heating evolves two gases (B) and (C) and an oxide (D). Gas (B) turns $K_2Cr_2O_7$ paper green while gas (C) forms a trimer in which there is no S-S bond. Compound (D) with HCl, forms a Lewis acid (E) which exists as a dimer. Compounds (A),(B),(C),(D) and (E) are respectively
- A) A,B,C are $FeSO_4, SO_2, SO_3$ B) D,E are $Fe_2O_3, FeCl_3$
 C) A,B,C are $SO_2, SO_3, AlCl_3$ D) D,E are $SO_3, AlCl_3$
44. Select the correct statements regarding the products obtained when $Na_2S_2O_3 \cdot 5H_2O$ is subjected to strong heating.
- A) Disproportionation reaction takes place
 B) Na_2SO_3 and Na_2S_5 are produced.
 C) Na_2SO_4 and Na_2S_5 are produced.
 D) One of the products has zig-zag open chain structure.
45. Which order is correct?
- A) $SF_6 > SeF_6 > TeF_6$: Order of stability
 B) $PCl_5 > SiCl_4 > AlCl_3$: Order of hydrolysis
 C) $H_2O > H_2S > H_2Se > H_2Te$: Order of thermal stability
 D) $SF_6 > SeF_6 > TeF_6$: Order of hydrolysis

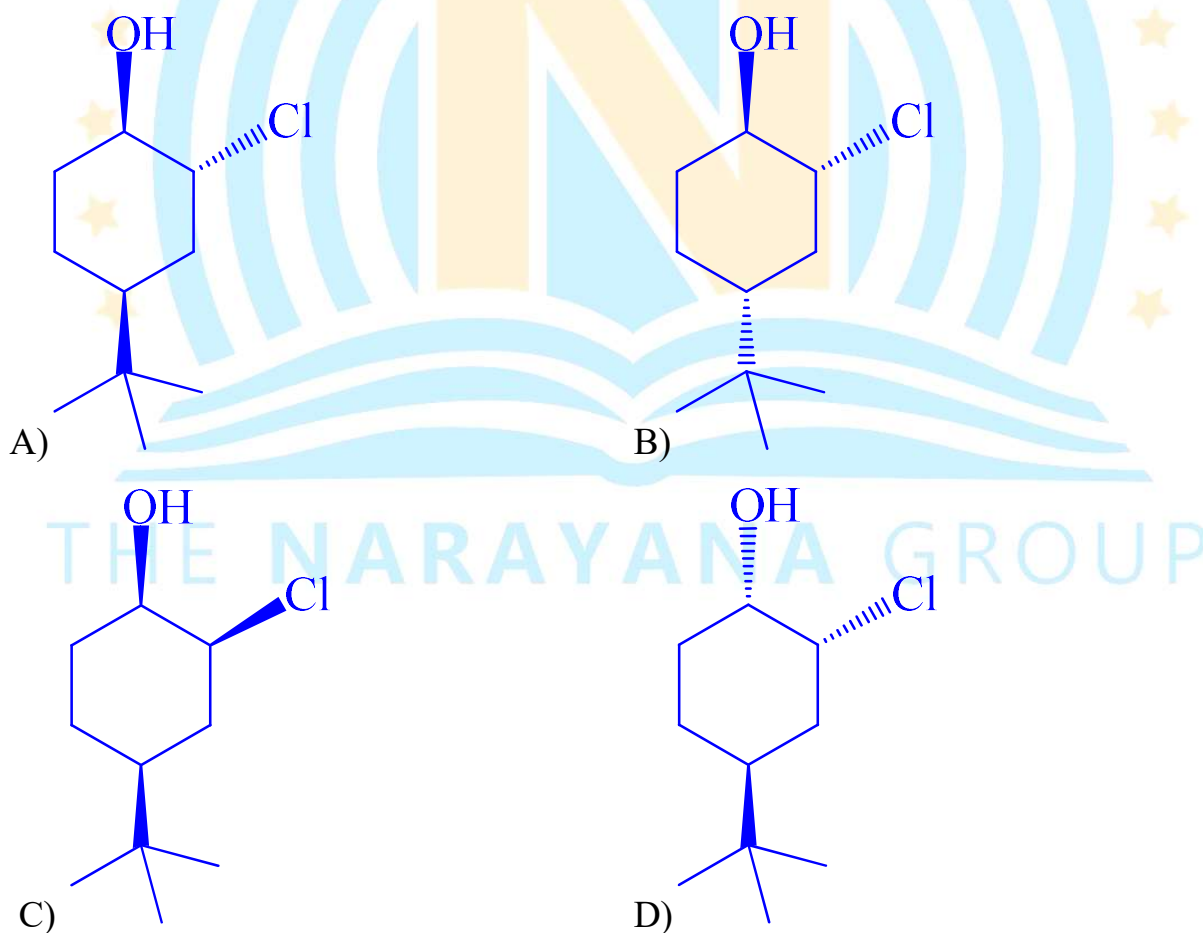
46. As Rhombic sulphur is heated in a test tube:

	Process	Temperature
a)	Viscosity increases	T_1
b)	Viscosity decreases	T_2
c)	Paramagnetic molecules	T_3
d)	Breakage of S_8 rings	T_4
	(Diradical formation in molten phase)	

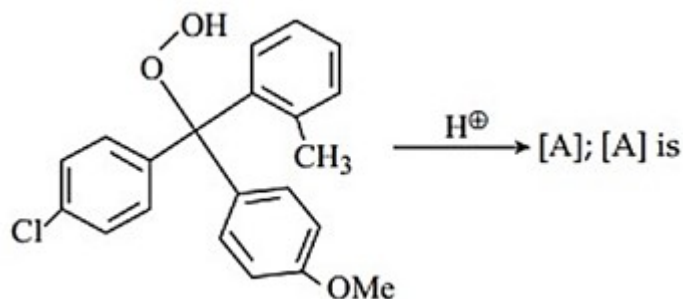
Correct order of temperature is:

- A) $T_4 < T_1$ B) $T_1 < T_2$ C) $T_2 > T_3$ D) $T_4 < T_3$

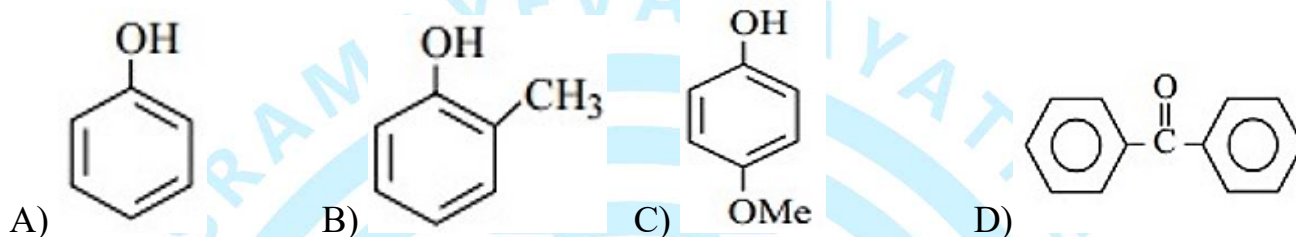
47. Which of the following will lead to the formation of epoxide?



48.



Which of the following is / are not the major product A?



SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

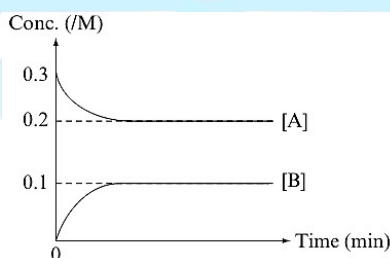
Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If **ONLY** the correct numerical value is entered as answer.

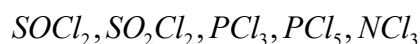
Zero Marks: 0 In all other cases.

49. For the reversible reaction $A(aq) \rightleftharpoons B(aq)$, which is first-order in both directions,

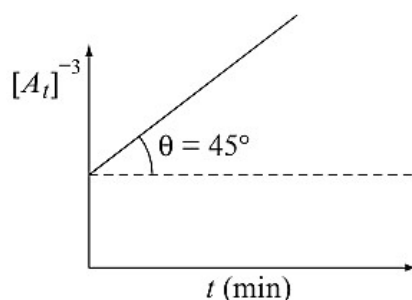
$K_f = \frac{1.38}{300} \text{ min}^{-1}$. The variation in concentrations with time shown in the figure.



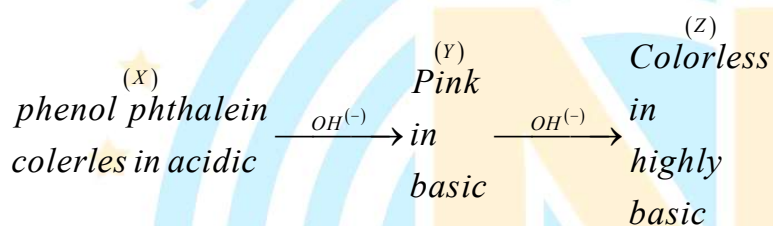
50. Complete hydrolysis product of 1 mole each of following will need how many total number of moles of NaOH for complete neutralisation?



51. For the reaction $3A \rightarrow 2B$, the following graph is obtained experimentally. The rate of reaction in milli molar sec^{-1} (mM sec^{-1}) when $[A] = 0.2M$, is



52. When phenol is treated with phthalic anhydride in presence of conc. H_2SO_4 phenolphthalein dye is formed. Which is used as a phthalein dye test for phenols. Phenolphthalein exists in different forms X,Y,Z at different pH values.



Sum of degree of unsaturation in these forms X, Y and Z is ...

53. Phenol reacts with bromine water to form 2,4,6-tribromophenol. If 94g of phenol react with excess of bromine to give 82.75g of 2,4,6-tribromophenol, calculate the percentage yield of the reaction. (atomic mass of C=12, H=1, O=16, Br = 80g/mol)
54. When salicylic acid is treated with excess of bromine water a white precipitate 'X' is formed degree of unsaturation in the compound 'X'